Kabir Singh

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Objective

Aspiring physician-scientist in translational medicine: designing peptide therapeutics computationally and modelling biological systems, alongside research in the history of medicine and early modern Sikh history.

Education

B.S. Mathematical Sciences (Mathematical Biology) Sep 2023 – May 2027 (Expected)
Albert Dorman Honors College and Jordan Hu College of Sciences and Liberal Arts, NJIT — Newark, NJ, USA

- Dean's List all semesters.
- Minors: Biomedical Engineering Technology, History, Philosophy and Applied Ethics; Honors tracks: Medical Humanities and Global Studies (first Honors College student to pursue two).
- Academic scholarships exceeding $100,000, including Academic Excellence, Honors Merit, Honors Residential, and Highlander Achievement scholarships; tuition and board scholarship support.

Biomedical	**Applications in Healthcare**; Biochemistry; Cell & Molecular Biology
Medicine & society	**Biomedical Ethics**; Health, Justice, & Capitalism; Medical Sociology; **Medicine & Law in Modern History**
Quantitative	Applied Numerical Methods; Differential Equations (ordinary & partial); **Linear Algebra**; **Mathematical Modeling**; Stochastic & Discrete Models in Biology
Independent study	Hawkes process model of amyloid nucleation (Dr James Maclaurin); **simplifying & modeling the MAPK/ERK pathway for therapeutic intervention** (Dr Horacio Rotstein, in progress); American medicine & end-of-life care (Dr Stephen Pemberton, in progress); intellectual history in the early modern Khalsa (Harvard Sikh Center & Satnam Singh, in progress)

Exchange semester (Global E3), Lund University, Sweden Jan 2025 – Jun 2025
Faculty of Engineering (LTH); 44 ECTS — Lund, Sweden

- Fellow, Institute of International Education's Global Engineering Education Exchange (Global E3).

Master's level	Biomathematics; Immunotechnology; Microscopy, Bio-Imaging; **Ultrasound Physics & Technology**
Undergraduate	Patent & Intellectual Property Rights

Research Experience

Computational and Mathematical Biology

Undergraduate Researcher and Computational Biology Team Lead Oct 2023 – Present
KumarLab; PI: Dr Vivek Kumar, University of Houston (previously NJIT) — Houston, TX / Newark, NJ, USA

- Lead undergraduate and graduate trainees in computational development of peptide therapeutics; trained 15+ lab associates and secured over $15,000 in direct project funding.
- Designed cyclic peptides against multiple sclerosis autoantibodies; ran molecular-dynamics simulations for the published lead peptide CAP8 (KPYPFWDD), from an eight-residue design scaffold and Rosetta/FlexPepDock workflow.
- Validated designs with 100 ns GROMACS simulations and alanine scanning; in the published study, CAP8 bound anti-GlialCAM autoantibody with $K_d = 0.294$ mM, versus 4.82 mM linear and 2.61 mM unrelated cyclic controls, and cut epitope detection on Western blot by 69% versus 27% and 5% for controls.
- Led a funded non-hormonal male-contraception project: machine-learning-guided cyclic peptide design against *redacted target*, docking and refinement; grant writing, manuscript and patent application in preparation.
- Conducted dissections and assisted animal work on angiogenic hydrogels subsequently published;

maintained computational research after the lab relocated to Houston.

Undergraduate Researcher, Computational Bionanotechnology Jun 2026 – Present
PI: Prof Stefano Angioletti-Uberti, Department of Materials, Imperial College London London, UK

- Apply statistical mechanics, molecular dynamics, protein language models and machine-learning methods to enzyme and binder design and soft-matter/biomaterial systems.
- Contributed code to BAGEL (Biomolecular Algorithm for Guidance in Energy Landscape), a tool for searching biomolecular energy landscapes; investigate peptide designs that mimic natural ligands.

Undergraduate Researcher, Systems Biology Sep 2026 – Present
PI: Dr Horacio Rotstein, NJIT Newark, NJ, USA

- Simplify and model the MAPK/ERK (Ras-Raf-MEK-ERK) signaling pathway in cancer as a dynamical system, to identify points for therapeutic intervention.

Undergraduate Researcher, Mathematical Modeling Oct 2025 – May 2026
PI: Dr James Maclaurin, NJIT Newark, NJ, USA

- Applied Hawkes self-exciting point processes to secondary nucleation and feedback in amyloid formation.

Intellectual and Cultural History

Emerging Scholars Research Fellow, Early Modern Sikh History Sep 2026 – Dec 2026
Harvard Sikh Center; mentor: Satnam Singh Remote

- Research 'Dissemination and the making of Khalsa authority in a contested panth': how the early Khalsa came to dominate the wider Sikh world against rival claimants, including the Bandai Khalsa, Minas, Ram Raiyas and Hindaliyas.
- Argue that the answer lies less in politics than in transmission: the Khalsa's consolidation of 'authentic Sikh teachings' through *katha* preceded and enabled its control of the panth.
- Sources: Gurmukhi and Persian primary materials, including works attributed to Bhai Mani Singh, *hukamname*, *rahit-nama* and sampardaic literature, alongside Persian and European accounts of the period.
- Twelve-week fellowship term producing a paper and a presentation: first, a characterization of the intellectual landscape and its conflicts; second, a prosopography linking intellectual influences to the period's political figures.

Undergraduate Researcher, History of Medicine and Bioethics Sep 2026 – Present
PI: Dr Stephen Pemberton, NJIT Newark, NJ, USA

- Trace the legal history of end-of-life decision-making from *In re Quinlan* (1976) to *Cruzan v. Director, Missouri Department of Health* (1990).
- Compare how three physician-authors, Sherwin Nuland, Atul Gawande and Haider Warraich, account for dying in modern American medicine; develop scholarship intended for submission to a peer-reviewed humanities journal.

Undergraduate Researcher, Digital History and Urban Studies Jan 2024 – May 2024,
PI: Dr Louis Hamilton, NJIT; field research in Rome and Florence Jan 2026 – May 2026
Newark, NJ, USA / Italy

- Two solo-authored geospatial narratives (StoryMapJS) mapping sites visited on foot against historical and archival sources.
- *Power vs. the People* (HIST 328): mapped 18 sites, layering Counter-Reformation Marian shrines against Julius II's 1508 papal axis along the Via Giulia, Rome, drawing on the Early Modern Rome Reconstructed database and Rufini's 1853 shrine census; argued the street shows superimposition and negotiation rather than a papal-versus-popular divide.
- *Processions and Power* (HIST 329): mapped 22 Florentine sites to examine how urban expansion transformed liturgical processions into contests of ecclesiastical and household prestige, read against Dante's critique of the *gente nuova*.

Emergency Care and Health Services

Undergraduate Researcher, Collegiate Emergency Medical Services Sep 2025 – Present
PI: Dr Chang Yaramothu, School of Applied Engineering and Technology, NJIT Newark, NJ, USA

- Co-developed IRB protocol, questionnaire and recruitment materials with two other students for a national study of collegiate EMS organizations; recruited via National Collegiate EMS Foundation regional coordinators.
- Examined resources, organizational structures, operating models and capabilities across approximately 50 US collegiate EMS squads; data analysis in progress.

Research Funding & Honours

Grants & awards

Undergraduate Research Opportunities Programme (UROP) bursary Jun – Jul 2026

- Imperial College London, Department of Materials; project: using protein language models to mimic natural ligands
- Personal compensation included

NSF I-Corps Regional Program, Northeast Hub May – Jun 2026

- Competitive cohort and customer-discovery project funding for Axioma Therapeutics

Grace Hopper AI Research Institute grant May – Aug 2025

- NJIT, computational protein design course development; $3,330

Honors College Moonshot Grant for Engaged Scholarship Jun – Jul 2025

- Male-contraceptive project; one recipient per annum
- Personal compensation included

Male Contraceptive Initiative Youth Advisory Board project grant Jan – Jun 2025

- Grantee-led male-contraceptive project
- Personal compensation included

Male Contraceptive Initiative travel award May 2025

- National Contraception Meeting, Durham, North Carolina

NJIT Undergraduate Research and Innovation Student Seed Grant Oct 2024 – May 2025

- Phase-2 longitudinal research project, cyclic peptide therapeutic for multiple sclerosis

Honors Summer Research Institute (HSRI) grant Jun – Jul 2024

- ADHC and NJIT URI; multiple sclerosis and male-contraceptive projects
- Personal compensation included

First place, Honors Interdisciplinary Research Forum Fall 2025

- Male-contraceptive poster, NJIT Albert Dorman Honors College

Academic Excellence, Honors Merit, Honors Residential scholarships 2023 – 2027

- NJIT merit scholarships, part of award package exceeding $100,000

Highlander Achievement Scholarship – NJIT 2023 – 2027

Fellowships & selective programmes

Emerging Scholars Research Fellowship 2026

- Harvard Sikh Center; early modern Sikh history

Honors Summer Research Institute Fellowship – Fellow Jun 2024 – Jul 2024, Jun 2025 – Jul 2025

Goldwater Ambassadors Program Aug 2025 – Present

- Selected mentee, mentored by Goldwater Scholars; programme for underserved or underrepresented students

NSF ART Integrated Student Ambassador Network (ARTISAN) Apr 2025 – Present

- Selected/inducted on the basis of translational-research engagement

No Patient Left Behind Fellowship – Fellow Jun – Aug 2024

American College of Cardiology – Young Scholar May 2023 – Apr 2024

PUBLICATIONS

[1] C. Heffernan, O. West, **K. Singh**, J. Dodd-o, M. James, A. Baradi, K. Rybak, A. Lateef, S. Elkabes, V. Kumar. "A novel cyclic peptide that blocks human autoimmune antibodies for the treatment of multiple sclerosis." *Eur. J. Med. Chem.* (2026). doi:10.1016/j.ejmech.2026.119173
Contribution: ran the molecular dynamics simulations; designed the published peptide

[2] **K. Singh**, C. Heffernan, O. West, J. Dodd-o, V. Kumar. "Machine learning for rational design of a cyclic peptide-based male contraceptive." In preparation; patent application in preparation

[3] J. Lau, D. Yun, **K. Singh**, A. Patananan, N. Jadavji. "Academic publishing as a launchpad: how publishing research in an undergraduate journal shapes future careers." In preparation

PRESENTATIONS

M. Kotta, G. Victorio, G. Karmacharya, **K. Singh**, N. Giovannetti, B. Shadpoor, J. Dodd-O, V. Kumar. "Putative computational design of novel inhibitors to modulate de-novo targets for CB1 receptor."
University of Houston, Undergraduate Research Day (*Poster and podium*) 2026

K. Singh, C. Heffernan, J. Dodd-O, V. Kumar. "Development of a novel male contraceptive using cyclic peptides to inhibit fertilization."
AAP/ASCI/APSA Joint Meeting (*Poster*) 2026
NJIT AI Exploration Day (*Poster*) 2026
NJIT Honors Interdisciplinary Research Forum, first place (*Poster*) Fall 2025
National Contraception Meeting (*Poster*) 2025
NJIT Honors Interdisciplinary Research Forum (*Poster*) Spring 2025
NJIT Dana Knox Research Showcase (*Poster*) 2025
NJIT Undergraduate Research and Innovation (*Abstract and podium*) 2025
Biomedical Engineering Society (BMES), co-presenter (*Abstract and poster*) 2024
NJIT Undergraduate Research and Innovation (*Abstract and podium*) 2024

K. Singh, J. Lau, H. Gerber, A. Alibilli, L. Goralsky. "Mitigating knowledge drain: a case study in optimizing workflow and training in a high-turnover, international peer-reviewed undergraduate science academic journal."
National Conference on Undergraduate Research (NCUR) (*Poster*) 2026

LECTURES & FEATURES

Invited speaker (forthcoming), Science National Honor Society – Piscataway High School 2026

Invited student roundtable speaker 2026
- Margo Chaly, New Jersey Secretary of Education; NJIT

Invited speaker, NJIT University Advancement Dinner 2026
- Interdisciplinary medicine and healthcare research for NJIT leadership, including interim president and interim provost

Speaker, 'Shaping the Future of Healthcare & Innovation' dinner Oct 2026
- Peptide therapeutics, published multiple sclerosis research, and emergency care; alongside Dr Genoa R. Warner and Dr Bryan Pfister, NJIT

Lecture series, Ethics in Science and Research – Science Mentorship Institute 2026

Session moderator, Student Oral Presentations (Communications) – NCUR 2026

Featured presenter, 'Navigating peer review and the student journal landscape' 2026
- 'Shaping Asia's Future: Development, Innovation, and Society in Transition', Nazarbayev University / Asian Universities Alliance

Featured panelist, 'It's Different for Girls' 2026
- Documentary screening and panel; NJIT ADHC and CSLA co-sponsored colloquium

NJIT Makes: Undergraduate Research and Innovation – NJIT promotional material 2026

Invited speaker, NJIT Board of Trustees and President's Town Hall 2025
- Presented with NJIT First Aid Squad leadership at the request of President Teik Lim

Honors in Action 2025 2025
- Albert Dorman Honors College publication: research, awards and study abroad

Article feature: '150+ Honors Scholars Showcase Findings at Interdisciplinary Research Forum' – Nayib Morán, NJIT News 2025

Featured speaker, ADHC Advising Day – NJIT; appearances in both 2024 and 2025 2024, 2025

Honors in Action 2024 2024
- Albert Dorman Honors College publication: research and study abroad

Article feature: 'Undergraduate Student Researchers on Campus' 2024
- Gurnoor Kaur, *The Vector*, NJIT student newspaper

Lecture, 'Introduction to PyMOL and Linear Peptide Design' – NJIT, Genetically Engineered Machines 2024

Clinical & Translational

Emergency medical services

Public Relations Chair; Emergency Medical Technician Oct 2025 – Present,
NJIT First Aid Squad Jan 2024 – Present
- Licensed New Jersey EMT; over 2,000 hours handling emergency medical situations across EMS roles.
- Represented the squad in presentations to NJIT senior leadership and Board of Trustees.

EMT Senior Squad Member; former Cadet Crew Chief, Marlboro First Aid Squad Apr 2022 – Jan 2025

Public health education

Stop the Bleed Instructor, NJIT; ACS-certified instruction Oct 2025 – Present

Stop the Bleed Instructor (hybrid English–Punjabi) Jan – May 2026
New Jersey gurdwaras; NJ Office of Homeland Security and Preparedness; Sikh Youth Alliance
- Taught hemorrhage-control skills to 100+ people across NJIT and gurdwara outreach.
- Bilingual delivery because those most likely to be first on scene in a gurdwara are not always the most comfortable in English.

Commercialisation

Co-founder and Technical Lead May 2026 – Present
Axioma Therapeutics; funded through NSF I-Corps Northeast Hub
- Led customer discovery and lean-startup validation for peptide and mini-protein therapeutics targeting degenerative eye diseases.

Academic Service

Editorial & peer review

22nd Editor-in-Chief; former Associate Editor Dec 2024 – Present
Journal of Young Investigators (JYI)

- Lead the oldest international undergraduate research journal, funded and featured by Nature, Science, NSF and the Burroughs Wellcome Fund; earlier associate-editor work spanned biology, physical sciences and social sciences.
- Developed structured, multi-stage hiring, asynchronous onboarding, tiered training, and a shared manuscript workflow board to reduce institutional knowledge loss.
- 2025–26: 27 new staff from 96 applications; two associate editors and a research editor assigned to each manuscript, with a two-week review window; author/reviewer AI policy published.
- Re-established translations, restored Perspective articles, and developed partnerships for undergraduate research events; convened at the Seattle executive-board summit.

Grant Reviewer, MCI Youth Advisory Board Undergraduate Project Grant Dec 2025 – Jan 2026

Teaching & mentorship

Research Lead Sep 2026 – May 2027
Governor's STEM Scholars; Research and Development Council of New Jersey (Expected)

- Mentor a high-school team on a year-long engineering and research project aligned to a UN Sustainable Development Goal and a New Jersey community need.
- Oversee meetings, budget, milestones and deliverables, with final presentation planned across four statewide conferences; $1,000 completion stipend.

Ethics Coordinator, Mentor (Neuroscience and Biology), Admissions Officer, Symposium Organizer and Grant Writer; former Ethics Curriculum Developer May 2024 – Present
Science Mentorship Institute (Sci-Mi)

- Contributed to ethics instruction and research mentoring; delivered 2026 lecture series on ethics in science and research.

Teaching Assistant, Honors First Year Seminar, Albert Dorman Honors College, NJIT Sep 2024 – Dec 2024, Sep 2025 – Dec 2025

Teaching Assistant, Computational Protein Design, Grace Hopper AI Research Institute, NJIT May – Aug 2025

Sikh History (Virsa) Teacher, Khalsa School, Sikh Sabha of New Jersey Sep 2020 – Present

Research-team mentor and computational instructor, KumarLab; 15+ trainees Oct 2023 – Present

National & professional

Research Chair Jun 2025 – Present
American Association of Physicians of Indian Origin (AAPI) PreMed Board

- Lead research-focused committee and programming to support students, researchers and physicians.
- Helped organize and host 2025 and 2026 Fall Virtual Research Symposia; sole MC in 2025 and co-MC in 2026, with 11 US and international presentations and seven physician judges recruited.

Board Member, Male Contraceptive Initiative (MCI) Youth Advisory Board Aug 2025 – Present

Member, Pre-Medical Ad Hoc Committee, American Physician Scientists Association (APSA) Mar 2026 – Present

University

Member, Student Conduct Board Dec 2024 – Present
NJIT

- Adjudicate academic and behavioral conduct cases alongside faculty, staff and university administrators.

Chair; former Chair-in-Training, Highlander Integrity Council, NJIT Sep 2024 – Present

Member, Inclusive Excellence Student Advisory Board, NJIT Feb 2025 – Present

Member, Multifaith Council, NJIT Jan 2025 – Present

Organizer, 'It's Different for Girls' screening and panel, NJIT ADHC and CSLA co-sponsored colloquium Mar 2026

University Representative, BioNJ 33rd Annual Dinner Meeting and Innovation Celebration Feb 2026

Steering Committee Lead, Albert Dorman Honors College First Year Retreat Mar – Aug 2025

Advisor; former President and Secretary, Honors Hall Council, NJIT Feb 2024 – May 2025

Honors College Representative, National Collegiate Honors Council Annual Conference Nov 2024

Leadership & Community

Vice President; former President and Events Coordinator Dec 2023 – Present
Sikh Association of NJIT (SANJIT)

- Vice President, 2026 – present; President, 2023 – 2026; earlier Events Coordinator.

Co-founder; former President and Treasurer, NJIT MedicZero Nov 2023 – May 2026

Co-founder, Delta Epsilon Mu, pre-medical professional fraternity Feb 2024 – May 2025

Youth Camp Organizer and Coordinator, Sikh Sabha of New Jersey Aug 2021 – Present

Youth Camp Counselor, Guru Nanak Sikh Heritage Aug 2021 – Aug 2025

Content Manager, The Mardana Project May 2026 – Present

Certifications

- New Jersey Emergency Medical Technician licence
- American College of Surgeons Stop the Bleed certification and instructor licence
- OSHA General Industry Safety and Health certification
- CITI Biomedical Research certification
- CITI Responsible Conduct of Research (RCR) Basic certification
- American Heart Association BLS and CPR, Adult/Child/Infant

Skills & Languages

Computational biology	Rosetta; RosettaScripts; PyMOL; GROMACS and GROMACS scripting; NAMD; VMD; OpenMM; CHARMM; AlphaFold; Boltz; ESMFold; protein language models
Methods	Computational peptide design; docking; molecular dynamics; statistical mechanics; RMSD analysis; mathematical and stochastic modeling
Laboratory	Surface plasmon resonance (SPR); general dissection
Data and tools	Python; LaTeX; molecular dynamics trajectory analysis
Languages	English (C2); Punjabi (C1); Swedish (B2); Spanish (B1); Russian (B1)

Interests

Gurmat Sangeet (classical Indian music in the Sikh devotional tradition); wrestling and martial arts; ping-pong; travel; reading history, epistemology and comparative theology.

References available upon request